

# Skills Summary

## Solving for Angle Measures on Parallel Lines

Use this reference while completing your worksheet or when studying.

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### Skill 1 — Congruent pairs: set the expressions equal

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On **parallel** lines cut by a transversal, three named pairs are *congruent*:

- **corresponding** — matching corners at the two intersections
- **alternate interior** — both between the lines, opposite sides of  $t$
- **alternate exterior** — both outside the lines, opposite sides of  $t$

Each of these gives the equation “first expression = second expression”. Substitute your answer back into both to check.

**Example:**  $j \parallel k$  and corresponding angles measure  $(3x + 7)^\circ$  and  $(5x - 23)^\circ$ . Find  $x$ .

**Step 1.** The pair is corresponding on parallel lines, so the angles are congruent — which turns the picture into an equality of the two expressions.

**Step 2.**  $3x + 7 = 5x - 23$ , so  $30 = 2x$ .

$$\Rightarrow x = 15$$

**Try it:** Alternate interior angles measure  $(2x + 18)^\circ$  and  $(4x - 6)^\circ$ . Find  $x$ .

check:  $21 = x$

### Skill 2 — Supplementary pairs: add to 180

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Two named pairs are *supplementary*:

- **same-side interior** — both between the lines, same side of  $t$  (needs parallel lines)
- **linear pair** — adjacent at one vertex, outer sides forming a line (needs nothing)

Each gives “first expression + second expression = 180”. Adding when the pair is congruent, or equating when it is supplementary, is the error this topic is built to catch.

**Example:**  $j \parallel k$  and same-side interior angles measure  $(5x + 12)^\circ$  and  $(3x + 8)^\circ$ . Find  $x$ .

**Step 1.** Both angles lie between the parallel lines on the same side of the transversal, so the pair is supplementary — the equation is a sum equal to 180, not an equality.

**Step 2.**  $(5x + 12) + (3x + 8) = 180$ , so  $8x + 20 = 180$  and  $8x = 160$ .

$$\Rightarrow x = 20$$

**Try it:** Same-side interior angles measure  $(4x + 9)^\circ$  and  $(2x + 15)^\circ$ . Find  $x$ .

check:  $92 = x$

### Skill 3 — Two-step chains

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When two angles are not a named pair, travel through a third. **Corresponding** or **vertical** moves you to the other intersection; a **linear pair** moves you around a vertex.

Track what each link does: congruent then congruent leaves the ends congruent; congruent then supplementary leaves the ends *supplementary*. Count the supplementary links — an odd number means subtract from 180.

**Example:**  $j \parallel k$  and one angle measures  $124^\circ$ . Find the angle reached by a corresponding step followed by a linear pair.

**Step 1.** The corresponding step copies the measure to the other intersection, and the linear pair then subtracts it from 180 — one supplementary link, so the answer is the supplement.

**Step 2.**  $180 - 124$ .

$\Rightarrow$  **56** degrees

**Try it:** The same chain starting from an angle of  $71^\circ$ . Find the measure at the end.

check: 601